

Guhan K B

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PROFESSIONAL SUMMARY

Final-year B.Tech student specializing in Artificial Intelligence and Data Science, with practical experience in Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), and Large Language Models (LLMs). Skilled in Python, PyTorch, Hugging Face, and modern AI development frameworks, with experience building end-to-end AI applications, Retrieval-Augmented Generation (RAG) systems, and computer vision solutions. Published researcher with four peer-reviewed IEEE and AIP papers, passionate about building scalable AI solutions to solve real-world challenges.

EDUCATION

Amrita School of Engineering, Coimbatore
B.Tech – Artificial Intelligence and Data Science

2023 – 2027
CGPA: 8.0/10

PROFESSIONAL EXPERIENCE

The Home Ideas

Oct. 2025 – April. 2026

Software Developer (AI/ML) · Remote

Bangalore

- Built a virtual interior design platform using **SegFormer** and **SAM 3** for automated wallpaper/flooring segmentation and visualisation.
- Implemented a **voice-enabled AI chatbot** with Gemini and OpenAI APIs; integrated **FAISS-powered retrieval** and web scraping for improved response accuracy.

HANUVEN HEALTHCARE

Jul. 2024 – Mar. 2025

Product Developer · Remote

- Built an **LLM-powered chatbot using Retrieval-Augmented Generation (RAG)** architecture, improving customer query resolution through context-aware responses generated from company data.

PROJECTS

Hybrid Lightweight YOLOv8/YOLOv10 for PCB Component Detection

- Achieved **0.887 mAP@0.5** across 22 PCB component classes using a lightweight hybrid YOLO architecture, reducing parameters by **46% in YOLOv10n** and **55% in YOLOv8n**

English-to-Tamil Abstractive Summarization & Translation | [GitHub](#)

- Developed an end-to-end **BART + IndicTrans2** NLP pipeline for abstractive summarization and neural machine translation. Fine-tuned on the **CNN/DailyMail** dataset; BART achieved **ROUGE-1: 0.538** and **ROUGE-L: 0.436**, outperforming PEGASUS (**ROUGE-1: 0.471**) on the same benchmark.

Hierarchical Control of Electro-Thermal Systems in Hospital Microgrids | [GitHub](#)

- Developed and compared **MPC and Fuzzy-PID** hierarchical control frameworks for hospital microgrid energy management; MPC achieved **50.8% cost reduction** vs. 11.1% for Fuzzy-PID over a Grid+PV baseline.
- Developed an end-to-end **BART + IndicTrans2** pipeline; evaluated using ROUGE, SacreBLEU, and chrF metrics.

Multimodal Voice Pathology Classification | [GitHub](#)

- Implemented a hierarchical multimodal framework achieving **82.93%** detection accuracy and **67.24%** diagnostic accuracy.

CVD Diagnosis and Correction App | [GitHub](#)

- Flutter app using **CNNs and GANs** to detect and correct Color Vision Deficiency with real-time analytics support.

SKILLS & TECHNOLOGIES

Programming: Python, C, MATLAB, MySQL, HTML, CSS

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG),

Frameworks & Libraries: PyTorch, Scikit-learn, Hugging Face Transformers, OpenCV, NumPy, Pandas, Matplotlib, FastAPI, Flask

Tools & Platforms: Git, MongoDB, FAISS, Ollama, Jupyter Notebook, CUDA,

Embedded & Edge AI: Raspberry Pi, Arduino,

RESEARCH & PUBLICATIONS

- Two and Four Wheels Differential Racing Bots** — IEEE [\[Link\]](#) Aug 2025
- Emotion-Aware Image Captioning using Deep Learning** — IEEE [\[Link\]](#) Mar 2026
- Bio-Inspired Snake Robot for Real-Time Victim Detection in Disaster Environments** - IEEE (**Best Paper Award**) [\[Link\]](#) Mar 2026
- Solar Powered Agricultural Robot for Irrigation** — AIP [\[Link\]](#) Feb 2026

AWARDS & LEADERSHIP

- Logistics Team Head**, Tensor (AI Dept. Club), ASE Coimbatore Aug. 2025 – Mar. 2026
- Member, Event Management**, Youth United Council of India Dec. 2024 – Jul. 2025